

Weaver Goldman

 github.com/We-Gold  weavergoldman.com  linkedin.com/in/weaver-goldman  we.goldm@gmail.com

EDUCATION

Worcester Polytechnic Institute

Projected May 2027

MS Data Science

GPA 4.0/4.0

BS Computer Science

Courses: MLOps, Machine Learning, Statistical Methods for ML, Database Management Systems, Algorithms

EXPERIENCE

AI Engineering Intern | Regeneron Pharmaceuticals, Inc.

May 2026 – Aug 2026

- Built an auditable AI/ML observability platform on AWS monitoring 30+ models, adopted by 3 pilot teams, and informed by 13 stakeholder interviews.
- Developed an environment-agnostic Python SDK for scalable data ingestion and model analysis.
- Tracked data subpopulations with an unstructured machine learning approach (GMM, BIC, and cost matching).
- Built a React dashboard to visualize model performance, data drift, and benchmarking results.
- Implemented an adaptive LLM benchmark system with over **60%** lower API costs than a naive evaluation strategy

Software Engineering Intern | Medtronic (Minimally Invasive Therapies Group)

June 2025 – August 2025

- Saved **400+** hours of manual work annually by automating surgical robot log analysis tasks with AI (LLMs).
- Developed an AI tool with RAG and MCP to analyze robot logs, performing expert-level analysis in minutes.
- Coordinated numerous AI agents and data sources with LangChain/LangGraph to perform complex tasks.

Web Development Lab Assistant | Laboratory of Spaceflight and Planetary Exploration

August 2024 – Present

- Under contract with **NASA**, developed a data-driven space mission planning web application.
- Optimized the database and graphing system to support **millions** of data points interactively.
- Led the development of an interactive, 3D Mars lander simulation.
- Built a robust CI/CD pipeline for the lab with Docker, unit testing, linting, and GitHub Actions.

Software Development and Research Intern | Gittlen Cancer Research Labs

May 2024 – August 2024

- Developed an Electron desktop application for extracting nerves and blood vessels from cloud-hosted medical scans, which reduces file sizes by up to **69,355%** (from 132.08 GB to 190.44 MB).
- Achieved approx. **4,000%** computation speed improvement with multithreading and multiprocessing.
- Utilized Docker to create a cross-platform Python backend and an extensible plugin system.

PROJECTS

Scaling Diffusion Models to Large Sparse Graphs | PyTorch, Python

August 2025 – May 2026

- 4-person senior project on efficient diffusion architectures for molecule and protein generation in drug discovery.
- Achieved linear space complexity for our diffusion process on large, sparse graphs.

#TeamGraduate AI | Python, FastAPI, React, WordPress, Docker, Google Cloud Run

Jan. 2026 – May 2026

- Developed a cloud-hosted LLM platform for generating curriculum-aligned (Namibian) educational content.
- Built AI authoring tools embedded in WordPress that reduced lesson creation time by **87.7%**.

Identifying Transit Accessibility Gaps in NYC | Scikit-Learn, Python, Plotly, GTFS

April 2025

- Identified underserved areas in NYC by using distance to the nearest subway station as a proxy for accessibility.
- Predicted weekly ridership revenue per location using a Random Forest Regressor ($R^2 = 0.675$), based on **6** years of data.
- Visualized results with interactive geospatial maps, highlighting areas of opportunity based on accessibility and predicted revenue metrics (potential stops predicted to produce **>\$4 million** in revenue annually).

SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C/C++, Java, $\text{\LaTeX}$

Libraries: PyTorch, LangChain, TensorFlow, PostgreSQL, React, Angular, Streamlit, Plotly Dash, Three.js

DevOps/MLOps: Prometheus, Grafana, Bash, Weights and Biases

Tools/Environments: Git/GitHub, Docker, Linux, GCP, Azure, AWS, Cloudflare, Figma, Jira